Tutor Aid – Final Presentation

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Introduction

Problem and Solution

As a tutor, I've noticed something:
Even when we *care* about our students and genuinely want to help, the process of simply *finding* students can be surprisingly difficult.

- Most tutors rely on word of mouth, which is slow and unpredictable.
- We don't have a professional space to show **who we are**; our teaching style, our personality, and why we connect well with certain learners.
- Parents and students often don't know how to choose the *right* tutor for their needs.
- On top of teaching, tutors end up managing messages, scheduling, cancellations, and admin on multiple apps.

Tutor Aid is a platform designed *from the tutor's perspective* because I've lived these challenges myself.

TutorAid:

- Helps tutors **present themselves professionally** with clear profiles, subjects, qualifications, and teaching style.
- Makes it easier for students and parents to **find tutors who genuinely fit their needs**, not just the first available person.
- Reduces admin by centralizing communication, scheduling, and updates.

The goal isn't just matching students and tutors, it's helping tutors grow with confidence, clarity, and purpose.

Why It Matters

TutorAid values the human side of learning.

- **For Tutors:**
  Confidence, visibility, and a sense of professionalism. Less burnout. More time to actually *teach*.

- **For Students:**
  Learning feels more personal, welcoming, and empowering; because they're guided by someone who truly fits their learning style.

- **For Parents:**
  Clarity, transparency, and peace of mind. They know progress, communication, and scheduling are all reliable and organized.

TutorAid isn't just a tutoring platform,
It's a community built to support growth, connection, and dignity in education.

Deployment

Cloud Platform

I decided to deploy my React app on Render and Vercel because of cost-restraints. GCP, Heroku, Azure and AWS turned out to be too complicated.

Vercel works fantastically with React applications.

Deployment Pipeline

Frontend - Vercel

- **Static assets** (HTML, CSS, JS)
- **Serverless rendering**
- Quick global CDN delivery

React builds into **static files**, so Vercel can:

- Serve it incredibly fast
- Auto-optimize images, caching, routing
- Deploy changes in seconds

Deployment Pipeline

Backend - Render

- **Long-running servers** (your Express app runs continuously)
- **Custom APIs**
- **Direct database connections**
- Environment variables + networking

Your backend needs to:

- Stay running to handle requests (Vercel shuts long-running processes down)
- Connect to your SQL database securely
- Manage sessions/auth logic

Deployment Pipeline

Database - Alwaysdata/PHPMyAdmin

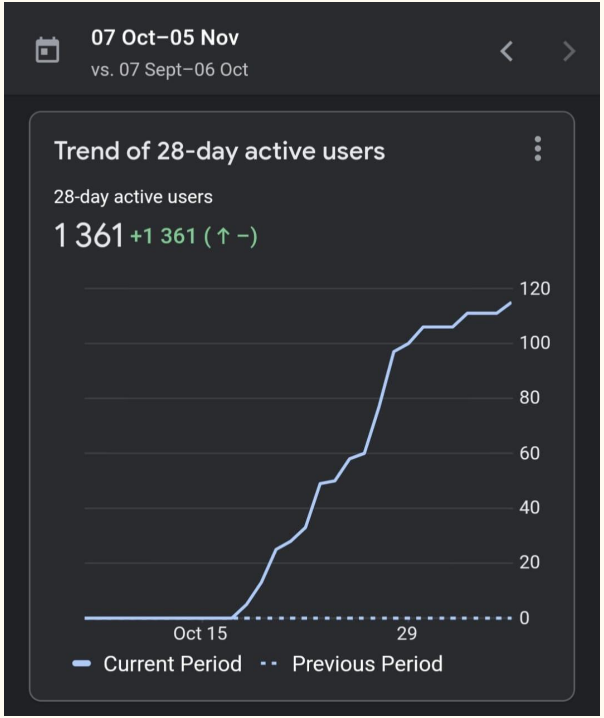
AlwaysData gives you:

- A **server**
- A **database** (MySQL or MariaDB)
- File hosting space

Where PHPMyAdmin is a management tool for your database.

SEO and Optimisation

Google Analytics



From the 15th of October to date, there has been an exponential increase of users. Let’s look at why this happened and if these are genuine users.

Google Analytics

A relief is that majority of the users are South African as they are my key audience.

1,361 active users in less than a month from **unexpected countries** because of a combination of:

1. **Domain name**
2. **Metadata / SEO keywords**
3. **How Google indexes new sites**
4. **Automated traffic + curiosity clicks**
5. **Hosting providers + deployment visibility**

Domain Name: gabydv.xyz

The **.xyz** top-level domain is:

- Popular globally
- Cheap, often used by tech communities and sometimes bots
- Frequently crawled by search engines and indexing platforms

This attracts **international automated traffic**, **not just South Africans**.

So some portion of:

- **China (82)**
- **Saudi Arabia (45)**
- **Germany (31)**

…are likely **crawler / bot / automated scanning traffic**, not real users.

This is **normal for any .xyz site**.

Metadata Broadly Targets General Tutoring Keywords

None of keywords restrict SEO to South Africa.

So Tutor Aid may appear in:

- Search results for **“Math tutor online”** in the U.S.
- Search results for **“Exam prep tutoring”** in Canada / UK
- Keyword scrapers in China that list service sites

Google **tests your site globally first**, then decides where to rank it long-term.

South African Traffic

826 South African users indicate **real local traction** which is great.

Your branding clearly signals:

- South African educational context
- Familiar subjects: EMS, Math Lit, AP Math, Afrikaans, Zulu
- Affordable hourly rates in **Rands**

Why So Much Growth So Quickly?

- **Launching a live public website, not just a portfolio build.**

Google rewards **sites with real user utility**.

- **Using simple, meaningful copy**

Google’s semantic crawler understands words like:
connects, *students*, *tutors*, *confidence* → that increases ranking.

- **The site has clear structure**

Your homepage is simple, meaning users **don’t leave immediately**, which boosts SEO ranking speed.

5. So Which Traffic Is “Real” vs. “Noise”?

So my **true audience** is **mostly South African**, which matches your goal.

Country	Type	Explanation
South Africa (826)	Real	Actual users / potential tutors / parents
United States (216)	Mixed	Some real students + some web crawlers
Canada / UK / France	Mostly Real	Searches for *online tutoring*
China, Saudi Arabia, Germany	Mostly Bots	Standard scanning of .xyz domains
“Not Set” (45)	Bots or VPN	API scans or anonymized browsing

Reflection and Conclusion

Reflection

This project was a full journey for me, not just in building a product, but in learning how to think like a full-stack developer.

One of the biggest learning curves was setting up and maintaining the SQL database and ensuring secure communication between the frontend and backend. I also had to learn how to deploy both sides of the app separately, which required research, patience, and **LOTS of trial-and-error**.

Through this, I really improved my debugging skills, my understanding of server-client communication, and how to optimise code for performance and loading speed.

Challenges and Solutions

One major challenge was deployment, managing the frontend, backend, and SQL database across different platforms. I solved this by hosting the backend on Render, the SQL database on AlwaysData, and using environment variables to securely connect everything.

I also ran into CORS issues when the frontend tried to call the backend, I resolved those by configuring allowed origins and carefully using environment variables.

Finally, I do not think I optimised performance by compressing images, using lazy loading for tutor images, and reducing unused CSS, like I should have.

What I learned this semester

I’ve gained a much stronger understanding of SQL, especially how to design structured relational data, write efficient queries, and maintain data integrity.

I also learned how to build and document APIs, handle routing and authentication, and manage state between frontend and backend.

Lastly, deployment taught me how important environment variables and secure configuration are, and how to think about real-world scalability and security.

Conclusion

This project means a lot to me because it comes from personal experience.
I built Tutor Aid to solve a real need, helping tutors find meaningful work and helping students find guidance, support, and confidence.

I’m proud of how far the project has come this semester, and I’m excited to keep developing it not just as a university assignment, but as something that can genuinely help people.